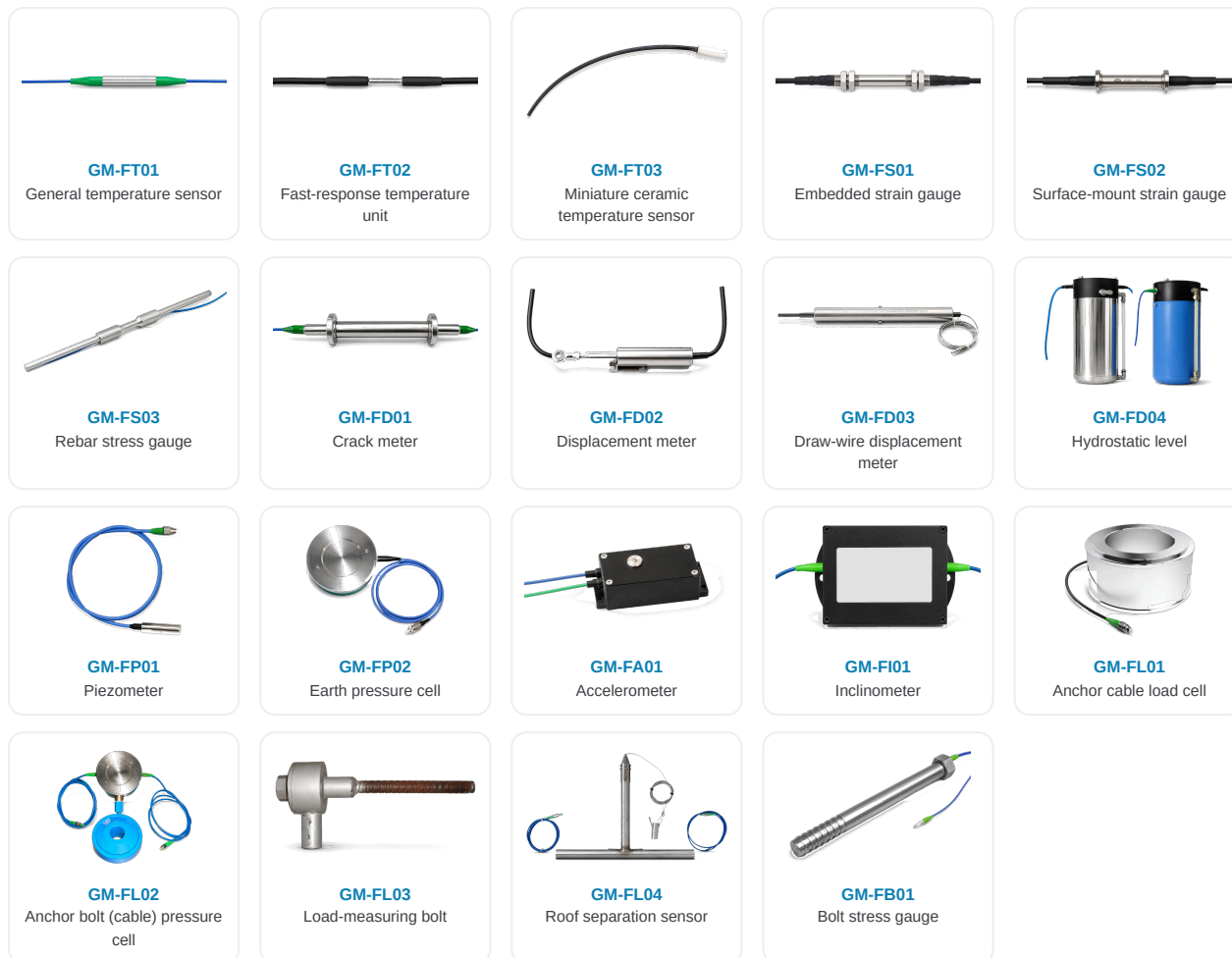


GM-F Series Fiber Bragg Grating Sensors

8 families · 21 standard models · IP68 across the range · C-band 1528~1568 nm · FC/APC

The GM-F series covers eight physical quantities: temperature, strain, displacement, pressure, vibration, inclination, anchoring and bolts. The sensing end is passive, EMI-immune and intrinsically safe; range, size and packaging are customizable across the range; strain types pair with GM-FT01 for temperature compensation; compatible with third-party C-band interrogators.

Product family



Naming & selection guide

Naming rule GM-F-T-01: GM brand · F grating sensor · quantity code · serial number. Quantity codes: T temperature / S strain / D displacement / P pressure / A dynamics / I inclination / L anchoring / B bolt. Three steps: ① pick the family by quantity; ② pick the model by range and mounting; ③ add one GM-FT01 per strain / displacement point for temperature compensation.

Platform parameters

Wavelength band	C-band 1528~1568 nm
Fiber connector	FC/APC
Ingress protection	IP68 across the range (see table for exceptions)

Interrogator compatibility	GM-FBG-16/32 and third-party C-band interrogators
Customization	Range / size / packaging / lead length per project

Specification table

Model	Name	Range	Accuracy / resolution	Key parameters
Temperature T · TEMPERATURE				
GM-FT01	General temperature sensor	-40~120 °C (custom)	±0.5 °C / 0.1 °C	Φ9×100 mm · IP68 · clamp / strap / embed
GM-FT02	Fast-response temperature unit	-40~120 °C (custom)	±0.5 °C / 0.1 °C	Φ7×100 mm · IP68 · hazardous areas · fire detection
GM-FT03	Miniature ceramic temperature sensor	-40~120 °C (custom)	±0.5 °C / 0.1 °C	Φ4×45 mm · IP68 · switchgear contacts, tight spots
Strain & rebar stress S · STRAIN				
GM-FS01	Embedded strain gauge	±1000 / ±1500 με	1% F.S. / 1 με	100 mm gauge · embedded · IP68 · pair with FT01 for temp compensation
GM-FS02	Surface-mount strain gauge	±500 / 1000 / 1500 με	1% F.S. / 1 με	100 mm gauge · weld on steel / bond on concrete · IP68
GM-FS03	Rebar stress gauge	200 MPa tension / 100 compression	1% F.S. / 1 MPa	All rebar sizes · 600 mm · butt-weld / strap
Displacement & settlement D · DISPLACEMENT				
GM-FD01	Crack meter	20 mm (custom)	1 / 0.1 mm	Fully sealed embedded · -40~80 °C · IP68
GM-FD02	Displacement meter	10 / 100 mm	0.1 / 0.5 mm · res. 0.01 mm	Φ40×200 mm · base mount, easy swap & calibration
GM-FD03	Draw-wire displacement meter	100 mm (custom)	1 / 0.1 mm	Φ40×350 mm · 20 m wire remote lead standard
GM-FD04	Hydrostatic level	0~100 mm	1 / 0.1 mm	Φ120×250 mm · networked settlement · stainless / molded
Pressure P · PRESSURE				
GM-FP01	Piezometer	0.35 / 1 MPa (custom)	0.5% / 0.1% F.S.	Φ30×120 mm · drop-in / embed · IP68
GM-FP02	Earth pressure cell	1 / 2 MPa (custom)	1% / 0.1% F.S.	Custom sizes · embedded · IP68 · pairs with piezometer
Dynamics A · DYNAMICS				

GM-FA01	Accelerometer	2 g · 0.5~50 Hz	2 mg / 0.5 mg · ≈500 pm/g	110×35×25 mm · IP68 · standard / enhanced · 2D / 3D combos
Inclination I · INCLINATION				
GM-FI01	Inclinometer	±5°	0.01° / 0.001°	105×80×28 mm · screw mount · IP68 · dual-grating self temp compensation
Anchoring & support L · ANCHORING				
GM-FL01	Anchor cable load cell (ex GM-PAC04)	0~2000 kN · 20 Hz	3~6 gratings redundant · 120° eccentric-load detection	-40~+85 °C · 95%RH · IP67
GM-FL02	Anchor bolt (cable) pressure cell	0~150 kN	5% F.S. · 1 kN	Sleeved between bolt plate and rock face · IP54
GM-FL03	Load-measuring bolt	70 kN	6% F.S. / 1% F.S.	Grating inside steel bolt · strain / axial force / bending / shear
GM-FL04	Roof separation sensor	200 mm, re-armable	5% F.S.	395×306×34 mm · early roof-separation trends
GM-FL05	Smart bolt	500 MPa tension	1% (0.5% optional)	FRP bolt body is the sensor · embedded grating · IP68
Bolt B · BOLT				
GM-FB01	Bolt stress gauge	100 MPa	<1% / ≤0.1% F.S.	Custom per bolt size · armored waterproof · -30~+80 °C

Wind-power products continue the proven supply models (GM-WTB / WTT / WTF series); see the Wind Turbine Monitoring Sensors datasheet.

Warranty & service

2 yr Hardware warranty · spares stocked	24 h On-site response, East China	48 h On-site response, nationwide	7×24 Remote diagnostics hotline
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Standard service policy; the supply contract prevails.

Certifications & standards

 China National Accreditation Service for Conformity Assessment CNAS testing	 计量认证 CMA metrology	 ISO 9001 quality	 ISO 14001 environmental	 矿用产品安全标志 MA mining safety
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Certificate numbers and scans are available on request.